

Page 3	Mark Scheme	Syllabus	Paper
	A LEVEL – NOVEMBER 2004	9702	4

(ii) $N = \{(2.24 \times 10^{-3})/224\} \times 6.02 \times 10^{23}$ 1
 $= 6.02 \times 10^{18}$ 1
activity = λN
 $= 2.23 \times 10^{-6} \times 6.02 \times 10^{18}$ 1
 $= 1.3 \times 10^{13} \text{ Bq}$ 1 **[4]**

(c) $A = A_0 e^{-\ln 2 \cdot t/T}$
 $0.1 = \exp(-\ln 2 \cdot n)$ 1
 $n = 3.32$ 1 **[2]**
(n = 3 without working scores 1 mark)